

CHEONKAM JEONG

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EDUCATION

The University of Arizona

Tucson, AZ

Ph.D. in Linguistics

2018–2024

- Dissertation: The Effects of Lexical Competition on Hyperarticulation of Voice Onset Time and Fundamental Frequency in Korean Stops: Experimental and Big Data Approaches (Advisor: Dr. Andrew Wedel)
- Research areas: Phonology, Computational Linguistics

The University of Arizona

Tucson, AZ

M.S. in Human Language Technology

2020–2021

- Internship Report: A semantic analysis of FindingFive support tickets using machine learning-based classifiers (Advisor: Dr. Andrew Wedel)
- Research areas: Natural Language Processing, Machine Learning

Hankuk (Korea) University of Foreign Studies (Hufs)

Seoul, South Korea

M.A. in English Linguistics

2015–2017

- Thesis: Noisy Harmonic Grammar Modeling of Flapping in American English Based on a Statistical Analysis (Advisor: Dr. Sung-Hoon Hong)
- Research areas: Phonology, Corpus Linguistics

Hankuk (Korea) University of Foreign Studies (Hufs)

Seoul, South Korea

B.A. in English Linguistics

2010–2015

- Research areas: English Linguistics, Philosophy

University of Leicester

Leicestershire, United Kingdom

Visiting Student Program

Fall 2013

- Department of English Literature (12th representative scholar sponsored by MIRAE ASSET, a South Korean financial services company)

WORK EXPERIENCE

Postdoctoral Scholar - AI/ML Lead in UC Noyce Project (PI: Dr. Adeline Nyamathi)

Sue & Bill Gross School of Nursing, University of California, Irvine

Irvine, United States

Jun 2024–Present

- Areas: Healthcare AI/ML, NLP, Reinforcement Learning

Visiting Scholar

Department of Computer Science, Purdue University

West Lafayette, United States

Jun 2024–Sep 2025

- Areas: NLP, Reinforcement Learning

NLP Intern

FindingFive (PI: Dr. Ting Qian)

New Jersey, United States

Jun 2020–Aug 2020

- Areas: NLP, Machine Learning

SKILLS

Languages: English, Korean, Japanese, French.

Programming: Python, R, SQL, Bash, Prolog, Perl

Speech/Audio: Praat/Parselmouth, Whisper, Wav2Vec2, OpenSmile, Kaldi

NLP/ML: PyTorch, HuggingFace Transformers, spaCy, NLTK, KoNLPy, LangChain

Infrastructure: Docker, PostgreSQL, GCP, Prodigy, WandB

Postdoctoral Scholar, AI/ML Lead

University of California, Irvine (PI: Dr. Adey Nyamathi)

Jun 2024–Present

- Designed empathy-focused conversational agents with evaluation metrics (Emotion Quotient; Cohen-Mansfield) for dementia patient care.
- Crafted and refined prompts conditioned on medical personas, personal stories, and prosodic cues for de-escalation responses based on expert-generated agitation scenarios.
- Harmonizing multimodal data (vision, audio, text) from real patient recordings; conducting qualitative analysis with clinical experts to inform model development.
- Collaborating with a robotics team to design and manage input/output pipelines for real-time robot interaction.
- **Pub:** Establishing the Foundations of Emotional Intelligence in Care Companion Robots. *JMIR Research Protocols*, 2024. [paper] [demo]
- **Pub:** DementiaBank-Emotion: A Multi-Rater Emotion Annotation Corpus for Alzheimer’s Disease Speech (Version 1.0). *HeaLing Workshop at EACL 2026* (Accepted).

Visiting Scholar

Purdue University, Department of Computer Science

Jun 2024–Sep 2025

- Developed LLM alignment with symbolic feedback using Direct Preference Optimization (DPO) for linguistic empathy.
- Operationalized linguistic empathy via Kuno-style perspective taking and dependency-parse validation; produced scalable preference pairs for alignment training, yielding 68.8% human preference gains ($p < .001$).
- Expanding the approach to counseling data.
- **Pub:** Parser-Validated Minimal-Pair Preferences for DPO: A Case Study in Legal “Empathy” (Submitted to *ACL 2026*). 📄

Research Associate (NLP Engineer / Developer / Phonetician)

The University of Arizona (Dept. of Linguistics; School of Information; Dept. of Computer Science; HLT; Douglass Phonetics Lab)

Spring 2021–2024

- **Dept. of Linguistics (Director: Dr. Gus Hahn-Powell)** Fall 2023
Built reproducible annotation infrastructure using Docker Compose and PostgreSQL; developed data pipelines and tooling to streamline IE annotation workflows.
- **School of Information (Directors: Dr. Mihai Surdeanu, Dr. Clayton Morrison)** Spring/Fall 2023
Annotated legal wills for IE (entities, binary relations, n -ary events); supported guideline/QC iterations (*Findings of EMNLP 2023*).
- **School of Information / Dept. of Computer Science (Director: Dr. Adarsh Pyarelal)** Spring/Summer 2023
Conducted multimodal multitask learning (text+audio); built LSTM/CNN/DistilBERT models and collaborated on unsupervised vocal entrainment detection (*Findings of NAACL 2025*); performed sentiment/emotion labeling and utterance-level IE for rule-based event extraction.
- **Dept. of Computer Science (NLP+CSS 2024)** 2024
Reviewed LLM-generated scenarios and labeled causal-question datasets; supported data-engineering pipelines (*NLP+CSS 2024*).
- **Human Language Technology (Directors: Dr. Sandiway Fong, Dr. Michael Hammond)** Spring 2023
Curated and archived HLT internship reports; performed keyword analyses to summarize themes and trends for program-level insights.
- **Douglass Phonetics Laboratory (Director: Dr. Natasha Warner)** Spring 2021
Designed and ran an online speech experiment (FindingFive; MTurk) with attention/quality checks; delivered reproducible cleaning and analysis notes.

NLP Engineer (Intern)
FindingFive, New Jersey

Jun–Aug 2020

- [Zoho Desk API] Built an automated pipeline to collect/clean and categorize support tickets; established basic analytics reporting.
- [Classical ML + Neural] Prototyped priority/triage classifiers (handoff/reassignment signals) and set baseline benchmarks.
- Closed the loop with operations via lightweight dashboards and feedback reports.

Data Study Group

The Alan Turing Institute, United Kingdom

Mar 2023

- Global Witness: Identifying unregulated mining sites from historical satellite data.
- Performed satellite image augmentation and quality control to improve data diversity and signal consistency.
- Trained and evaluated a Random Forest baseline over temporal composites to flag likely unregulated sites.
- **Pub:** Data Study Group final report: Global Witness. *Zenodo*, 2023. [\[paper\]](#)

Korean Education Dataset for Learners of Korean as a Foreign Language

Ministry of Science and ICT & National Information Society Agency (Korea)
2022–May 2023

Oct

- Audited and corrected prosodic/phonological tags; established reviewer checklists for consistency.
- Evaluated learner productions and documented recurrent error patterns to inform rubric refinements.

PUBLICATIONS

1. Jeong, C., & Wedel, A. (2026). The effect of contrast-specific minimal pair competitor in hyperarticulation of VOT and F0 phonetic cues in Korean initial stops in tonogenetic sound change. *Journal of Phonetics*, 115, 101474. <https://doi.org/10.1016/j.wocn.2026.101474>
2. Jeong, C., Liao, J., Lu, A., Song, Y., Rashidian, C., Krogh, D., Krogh, E., Rasouli, M., Lee, J.-A., Dutt, N., Gibbs, L. M., Sultzer, D., Rousseau, J., Ludlow, J., Galvez, M., Nuth, A., Khay, C., Brunswicker, S., & Nyamathi, A. (Accepted). DementiaBank-Emotion: A Multi-Rater Emotion Annotation Corpus for Dementia's Speech (Version 1.0). In *Proceedings of the 1st Workshop on Linguistic Analysis for Health (HeaLing) at the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*, Rabat, Morocco. Association for Computational Linguistics (ACL). <https://arxiv.org/abs/2602.04247>
3. Pyarelal, A., Culnan, J. M., Qamar, A., Krishnaswamy, M., Wang, Y., Jeong, C., Chen, C., Miah, M. M., Hormozi, S., Tong, J., & Huang, R. (2025). MultiCAT: Multimodal communication annotations for teams. In *Proceedings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025)*. Association for Computational Linguistics (ACL). <https://aclanthology.org/2025.findings-naacl.61/>
4. Nyamathi, A., Dutt, N., Lee, J.-A., Rahmani, A. M., Rasouli, M., Krogh, D., Krogh, E., Sultzer, D., Rashid, H., Liqat, H., Jawad, R., Azhar, F., Ahmad, A., Qamar, B., Bhatti, T. Y., Khay, C., Ludlow, J., Gibbs, L., Rousseau, J., Abbasian, M., Song, Y., Jeong, C., & Brunswicker, S. (2024). Establishing the foundations of emotional intelligence in care companion robots to mitigate agitation among high-risk patients with dementia: Protocol for an empathetic patient-robot interaction study. *JMIR Research Protocols*, 13, e55761. <https://doi.org/10.2196/55761>
5. Dumitru, R., Alexeeva, M., Alcock, K., Ludgate, N., Jeong, C., Abdurahaman, Z. F., Puri, P., Krichhoff, B., Sadhu, S., & Surdeanu, M. (2024). Retrieval augmented generation of subjective explanations for socioeconomic scenarios. In *Proceedings of the Sixth Workshop on Natural Language Processing and Computational Social Science (NAACL NLP+CSS 2024)*, pages 68–85, Mexico City, Mexico. Association for Computational Linguistics (ACL). <https://aclanthology.org/2024.nlpcss-1.6/>
6. Jeong, C., & Wedel, A. (2024). Applying big data and automation techniques in phonetics: A case study

on hyperarticulation in Korean word-initial stops. In *Proceedings of the 19th Conference on Laboratory Phonology (LabPhon 19)*. Hanyang University, Seoul, South Korea. <https://labphon.org/sites/default/files/labphon19/Papers/corpusphon/jeong%20and%20wedel.pdf>

7. Kwak, A., Jeong, C., Vincent Forte, G., Bambauer, D., Morrison, C. T., & Surdeanu, M. (2023). Information extraction from legal wills: How well does GPT-4 do? In *Findings of the Association for Computational Linguistics (EMNLP 2023)*, Singapore. Association for Computational Linguistics (ACL). <https://aclanthology.org/2023.findings-emnlp.287/>
8. Jeong, C., Schmitz, D., Kakolu Ramarao, A., Stein, A., & Tang, K. (2023). Linear discriminative learning: A competitive non-neural baseline for morphological inflection. In *Proceedings of the 20th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology*, pages 138-150, Toronto, Canada. Association for Computational Linguistics (ACL). <https://aclanthology.org/2023.sigmorphon-1.16/>
9. Jeong, C., & Wedel, A. (2023). The effect of cue-specific lexical competitors on hyperarticulation of VOT and F0 contrasts in Korean stops. In *Proceedings of the 2022 Annual Meeting on Phonology (AMP)*, Los Angeles, California. <https://journals.linguisticsociety.org/proceedings/index.php/amphonology/article/view/5438/5230>
10. Jeong, C., & Wedel, A. (2023). The effect of cue-specific lexical competitors on hyperarticulation of VOT and F0 contrasts in Korean stops. Poster presented at *Hanyang International Symposium on Phonetics and Cognitive Sciences of Language 2023 (HISPhonCog 2023)*. Hanyang University, Seoul, South Korea.
11. Data Study Group Team. (2023). Data Study Group final report: Global Witness - Identifying unregulated mining sites using historical satellite data (Version 1). *Zenodo*. <https://doi.org/10.5281/zenodo.8366048>
12. Jeong, C., & Park, S. (2018). A corpus-based study on the prosodic features of *com* in Korean. Poster presented at *92nd Annual Meeting of Linguistic Society of America (LSA)*, Salt Lake City, Utah. <https://www.lsadc.org/files/LSA%202018%20Annual%20Meeting%20Handbook.pdf>
13. Jeong, C., & Park, S. (2017). The role of prosody in discourse: A case study of Korean *com*. In *Proceedings of 2017 Seoul International Conference on Speech Sciences (SICSS)*, pages 109-110, Seoul, South Korea. https://www.academia.edu/35282293/The_role_of_prosody_in_discourse_A_case_study_of_Korean_com

MANUSCRIPTS

1. Jeong, C., & Nyamathi, A. (2026). Discourse position and context length matter: Information-theoretic analysis of text-based emotion in IEMOCAP. *arXiv preprint arXiv:2601.00181*. <https://arxiv.org/abs/2601.00181>
2. Nyamathi, A., Rasouli, E., Jeong, C., Brunswicker, S., Agapie, E., Ludlow, D., Jawad, A., Khay, N., Lu, L., Ding, C., Liao, F., Farahani, F., Lee, J., Dutt, A., Rahmani, A., & Krogh, P. (Under review). Evaluation of personas and daily activity maps for improved clinical care and training models for effective human-robot interaction for persons with dementia: A qualitative study. *Journal of Medical Internet Research (JMIR)*. <https://preprints.jmir.org/preprint/85651>
3. Jeong, C., Kim, S., & Park, J. (2025). The algebra of meaning: Why machines need Montague more than Moore's Law. *arXiv preprint arXiv:2510.06559*. <https://arxiv.org/abs/2510.06559>
4. Kwak, A., Jeong, C., Lim, J., & Min, B. (2024). A Korean legal judgment prediction dataset for insurance disputes. *arXiv preprint arXiv:2401.14654*. <https://arxiv.org/abs/2401.14654>
5. Yoo, Y., Jeong, C., Gim, S., Lee, J., Schimeke, Z., & Seo, D. (2023). A novel patent similarity measurement methodology: Semantic distance and technological distance. *arXiv preprint arXiv:2303.16767*. <https://doi.org/10.48550/arXiv.2303.16767>

INVITED TALKS	Jeong, C. (2025, May 20). Introduction to NLP and its Applications in Healthcare. <i>Sue & Bill Gross School of Nursing, University of California, Irvine</i> . Irvine, California, United States. Slides	
	Jeong, C., Darling, M., & Dely, A. (2023). Enterprise AI: How are you addressing the new wave of computing? <i>Tucson's IT Networking Group LAMBDA</i> . Tucson, Arizona, United States.	
	Jeong, C., & Romero, D. (2022). Student internships in industry: Experiences and resources. <i>The Arizona Research Bazaar 2022</i> . Tucson, Arizona, United States. Slides	
	Jeong, C. (2022). 2020 NLP internship at FindingFive. <i>Master of Science in Human Language Technology</i> , The University of Arizona. Tucson, Arizona, United States.	
TEACHING EXPERIENCE	Instructor Department of Linguistics	2019–2021
	• LING 341: Language Development (Psycholinguistics) • LING 314: Phonetics	Winter 2021 Spring 2020, Fall 2019, Fall 2020
	Teaching Assistant/Associate Department of Linguistics & College of Social and Behavioral Sciences	2018–2022
	• SBS 200: Introduction to Statistics for the Social Sciences • LING 315: Introduction to Phonology • LING 341: Language Development • LING 150: Language in the World	Spring 2022, Fall 2021 Spring 2019 Spring 2019 Fall 2018
RESEARCH PROPOSALS AND FUNDING APPLICATIONS	• OpenAI AI and Mental Health Grant Program (PI) • UCI MIND ADRC Developmental Grant (Co-PI) • NIH NIBIB Bioengineering Partnerships with Industry, U01 (Co-PI) Preparation • NVIDIA Saturn Cloud Academic Grant (Key Personnel) • AlzInsights AI Prize (PI) • Meta Large Language Model Evaluation Research Grant (PI)	2026, Submitted 2026, Not Funded 2026, In 2025, Awarded 2025, Not Funded 2024, Not Funded
AWARDS AND CERTIFICATES	Computational Social Sciences Certificate	
	<i>College of Social & Behavioral Sciences, The University of Arizona</i>	Spring 2022
	Certificate in Essay Writing	
	<i>Dept. of Philosophy, HUFS</i>	2015
	MIRAE ASSET Student Exchange Scholarship	
	<i>MIRAE ASSET, South Korea</i>	2013
	Second Distinction (Dean's recognition)	
	<i>Dept. of English Linguistics, HUFS</i>	2011

ACADEMIC SERVICES AND VOLUNTEER- ING

Paper Reviewer for:

- IEEE Access: Specializing in NLP Papers July 2023–Present
- Arizona Linguistics Circle 16: Phonetics & Statistical Modeling Proceedings Reviewer 2022
- Journal of Phonetics: Co-reviewer of PHONETICS-D-21-00128 with Dr. Andrew Wedel 2022
- Arizona Linguistics Circle 14, 15: Phonology & Phonetics Coyote Paper Reviewer 2021–2022
- Arizona Linguistics Circle 15: Phonetics & Corpus Linguistics Proceedings Reviewer 2021
- Arizona Linguistics Circle 14: Phonology & Phonetics Proceedings Reviewer 2020
- Arizona Linguistics Circle 13: Phonology Proceedings Reviewer 2019

Open Educational Resources:

- Modular Linguistics (with Dr. Andrew Carnie), University of Arizona Open Textbooks In Progress

Leadership and Communication:

- NAVER Boost Course: Coaching Study AI Basics 2023 - Team Leader January 2023
- FindingFive: Outreach and Marketing Coordinator March–August 2022
- Arizona CatCloud Service: Helped design the service with fellow student leaders.
- Arizona Linguistics Circle 15: Communication Coordinator 2021
- Arizona Linguistics Circle 13: Phonetics Section Chair 2019
- Education Supporter at Seoul Urban Science Technical High School in Seoul, South Korea 2013

REFERENCES

Research Advisors and Collaborators

- **Dr. Andrew Wedel** — Professor, Linguistics, The University of Arizona
wedel@arizona.edu
- **Dr. Adeline Nyamathi** — Distinguished Professor and Founding Dean, School of Nursing, University of California, Irvine
anyamath@uci.edu
- **Dr. Michael Hammond** — Professor, Linguistics & HLT, The University of Arizona
hammond@arizona.edu
- **Dr. Natasha Warner** — Professor & Dept. Head; Director, Douglass Phonetics Lab, The University of Arizona
nwarner@arizona.edu
- **Dr. Miquel Simonet** — Professor, Spanish & Portuguese; Cognitive Science, The University of Arizona
simonet@arizona.edu
- **Dr. Sandiway Fong** — Associate Professor, Linguistics & Computer Science, The University of Arizona
sandiway@arizona.edu
- **Dr. Gus Hahn-Powell** — Assistant Professor, Linguistics; Director, HLT Online MS & NLP Certificate, The University of Arizona
hahnpowell@arizona.edu
- **Dr. Mihai Surdeanu** — Associate Professor, Computer Science, The University of Arizona
msurdeanu@arizona.edu
- **Dr. Clayton Morrison** — Associate Professor, School of Information, The University of Arizona
claytonm@arizona.edu

- **Dr. Adarsh Pyarelal** — Assistant Professor, School of Information, The University of Arizona
adarsh@arizona.edu
- **Dr. Kevin Tang** — University Professor, English Linguistics, Heinrich Heine University Düsseldorf (Germany)
kevin.tang@hhu.de

Industry Collaborators

- **Dr. Ting Qian** — President; VP of Engineering, FindingFive
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Teaching References

- **Dr. Cecile McKee** — Professor, Linguistics, The University of Arizona
mckee@arizona.edu
- **Dr. Diane Ohala** — Associate Professor of Practice, Linguistics, The University of Arizona
ohalad@arizona.edu
- **Dr. Amy Fountain** — Associate Professor of Practice, Linguistics, The University of Arizona
avf@arizona.edu
- **Dr. Suzanne Delaney** — Associate Professor of Practice, Social & Behavioral Sciences, The University of Arizona
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